

Lens opacity mechanisms

TYPE B | Protein order, fibre hydration and residual epithelium.

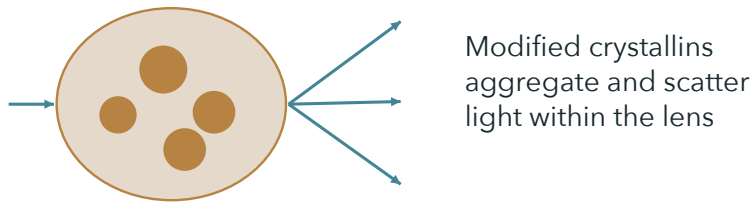
Transparency is an organised material state

Clear lens versus scattering lens

CLEAR: ORGANISED CRYSTALLINS



CATARACT: DAMAGED PROTEIN ORDER



Scattering reduces image quality even at the right focus

Normal concentrated crystallins remain stably organised. Damage and aggregation disrupt transmission; this opacity mechanism is different from presbyopia.

Locate the cataract pattern

Pattern	High-yield association / effect
Nuclear	Protein/fibre changes; may cause index myopia
Cortical	Hydration changes produce radial spoke-like opacities
Posterior subcapsular	Central posterior opacity; glare; steroid association

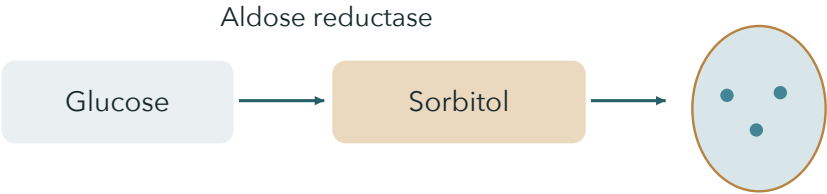
Three different explanations for near blur

Presbyopia reflects reduced accommodation. Nuclear index changes may temporarily improve unaided near vision without restoring accommodation. Cataract light scattering reduces image quality even if focal power is adequate.

Osmotic swelling versus capsular clouding

Two mechanisms, two tissue compartments

DIABETIC NATURAL LENS



Retained polyol -> water influx -> fibre swelling

AFTER NATURAL LENS EXTRACTION



Capsule clouding; artificial optic can remain clear

The polyol pathway can contribute to diabetic lens injury. Posterior capsule opacity is residual-cell growth after extraction, not a new cataract inside a clear artificial lens.

Trigger → material change

Trigger	Mechanism
Sustained excess lens glucose	Aldose reductase → retained sorbitol → osmotic water influx
Ageing / oxidative stress	Crystallin damage and aggregation increase scattering
Retained epithelial cells after extraction	Migration, proliferation and fibrosis can cloud posterior capsule

Keep the compartment explicit

Lens-fibre swelling occurs in natural lens tissue. Posterior capsule opacity lies on a retained capsule behind an artificial optic. Neither lesion is retinal fluid or a corneal endothelial infiltrate.